
APS360 ASL Alphabet Translator Project Proposal

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Google Colab Notebook GitHub Repo

Introduction

As someone who has witnessed first-hand how communication barriers can isolate individuals especially when travelling to non-English speaking countries, I am drawn to building technology that bridges the gap between the Deaf and hard-of-hearing community, and the non-signing majority. American Sign Language(ASL) is the primary language of over 500, 000 people in North America, yet the majority of the population cannot understand it. Human interpreters are expensive, scarce and often not available for everyday interactions. This project fills the gap by building a real-time ASL Alphabet Translator that classifies static hand signs into their corresponding English letters (A-Z) using webcam imagery. Deep learning is the ideal approach for this project because hand gestures are highly dimensional, non-linear spatial patterns that vary across hand size, lighting, skin tone and camera angle which traditional programming cannot account for. However, Convolutional Neural Networks(CNNs) are specifically designed to learn translation-invariant spatial hierarchies from raw pixel data making them well-suited to extract the finger shape and configuration features that distinguish one ASL letter from another. Success in this project would demonstrate a low cost, accessible alternative to interpreter services for everyday communication.

Illustration

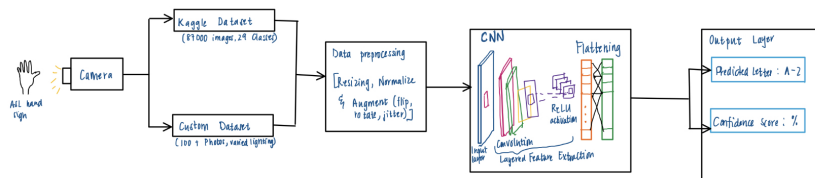


Figure 1: ASL Alphabet Translator CNN Pipeline.

Background & Related Work

This project builds on existing computer vision research and accessibility software in the gesture-recognition space. Google's MediaPipe Hands framework maps 21 3D landmarks of a hand from a single frame, representing the industry standard for hand tracking(1). The ASL Alphabet Kaggle Dataset contains 87,000 labeled images of ASL letters across 29 classes, which serves as the primary benchmark for training(2). Academic work on transfer learning shows that pre-trained architectures such as VGG16 and ResNet can be effectively fine-tuned for hand-shape classification(3). Commercial products like SignAll rely on multi-camera arrays and depth sensors to translate sign language, highlighting the need for simpler, single-webcam software solutions(4). Lastly, the APS360 Lab 3 demonstrates that a basic CNN can successfully differentiate between 9 distinct hand gestures using localized pixel convolutions thus providing a direct foundation for this project.(5).

Data Processing

The primary training dataset will be the publicly available ASL Alphabet Dataset from Kaggle, which provides 200×200×3 RGB images across 29 classes (A-Z, space, delete, nothing). Additionally, to

prove model robustness and generalization, I will collect a custom test dataset of over 100 original photos of my own hands performing each sign across varied room lighting, cluttered backgrounds, and different webcam angles. All images will be resized to a uniform tensor dimension of $224 \times 224 \times 3$ and normalized with standard ImageNet mean and standard deviation values. To minimise overfitting to the clean Kaggle data, I will apply random horizontal flipping, rotation up to 15 degrees and color jittering to simulate changing ambient room lighting.

Architecture

The model will be a custom CNN for 26-class image classification. Building on my lab experience, I will design a custom architecture featuring three to four sequential convolutional blocks. Each block will contain a Conv2d layer to extract edge and shape features of the fingers, a BatchNorm2d layer for training stabilization, a ReLU activation function, and a MaxPool2d layer to reduce spatial dimensions. The flattened output will feed into fully connected linear layers, ending in a 26-neuron output layer (mapping to letters A-Z). Optimization will be handled via the Cross-Entropy Loss function.

Baseline Model

To verify the deep learning approach, I will compare the CNN against a classical machine learning baseline. I will extract 1D Histogram of Oriented Gradients (HOG) feature vectors from the raw images to capture basic hand edge structures. These HOG vectors will be fed into a standard Support Vector Machine (SVM) classifier. Success for this project will be defined as the CNN achieving a test accuracy that significantly outperforms this non-deep-learning SVM baseline, particularly on the custom, messy-background test dataset.

Ethical Considerations

The primary ethical risk is representation bias. If the model is trained exclusively on the Kaggle dataset which may predominantly feature lighter skin tones or perfect hand dexterity, the network may systematically fail for users with darker skin tones or limited hand mobility like individuals with arthritis who cannot form perfect hand shapes. If relied upon in critical environments such as in medical or legal settings, this failure could actively marginalize the very community it is designed to assist. To mitigate this bias, my custom data collection must intentionally include varying skin tones and non-ideal hand formations.

References

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- [3] L. Pigou, S. Dieleman, P.-J. Kindermans, and B. Schrauwen, "Sign Language Recognition Using Convolutional Neural Networks," Springer International Publishing. Accessed: June 04, 2026. [Online]. Available: <https://link.springer.com/chapter/10.1007/978-3-319-16178-540>
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